GENESIS
IDEATHON

TEAM & DOMAIN

- Team Name –Aether
- Team Members –Adithya Arun:CB.SC.U4AIE26002
  Aagney Kanjur Vinod:CB.SC.U4AIE26136
  Nandana Arun Nair:CB.AI.U4AAR26037
  Pallavi S Nair:CB.AI.U4AAR26040
- Selected Domain – 4
- Problem Statement – How can we make e-waste recycling in India as easy and rewarding as selling an old device, while ensuring it reaches authorized recyclers safely?



TEAM AETHER

DOMAIN PROBLEM IN OUR WORDS

GENESIS

Explain the problem in your domain simply, in your own words

- What exactly needs to be solved?
- Ans-India is one of the largest producers of e-waste, but over 90% of it is handled by the informal sector, who burn electronics to extract metals, causing massive toxic pollution. Most people keep old tech in drawers because recycling it is too inconvenient.
- Why is it important right now?
- Ans-It is important now because the amount of electronic waste in India is growing rapidly as people replace phones, laptops, and other devices more frequently. Many old devices are simply stored at home or sold through informal channels, where unsafe recycling can cause pollution and health risks. A convenient system that gives people instant rewards for responsibly recycling their electronics can increase participation while also recovering valuable materials.

TEAM AETHER

WHO IS AFFECTED?

GENESIS

Target users, stakeholders, beneficiaries

- Primary users / target group: People who have old or unused phones, laptops, tablets, chargers, and other electronic devices and want an easy way to dispose of or recycle them.
- Secondary stakeholders: Authorized e-waste recyclers and collection partners, device manufacturers and producer responsibility organisations under EPR rules, government and pollution control regulators, and informal collectors who could be brought into the formal system as pickup partners.
- Who benefits if the problem is solved?
- Ans-Consumers benefit by getting instant rewards for their old electronics, while recyclers get more organized e-waste. The environment and society benefit through reduced pollution, safer recycling, and recovery of valuable materials.

CURRENT SITUATION & CONTEXT

Existing scenario, background and how things work today

- Growing electronics use is increasing e-waste in India.
- Many people store old devices because disposal is inconvenient.
- Informal collectors are commonly used for unwanted electronics.
- Device value and recycling options are often unclear.
- India generated 13.98 lakh tonnes of e-waste in FY 2024–25.
- 28.75 lakh tonnes were recycled through the EPR system in FY 2025–26.
- 386 registered e-waste recyclers were reported as of March 2026.
- Current process: Unused Device → Find Buyer/Recycler → Valuation → Pickup/Drop-off → Sorting → Recycling/Refurbishment → Payment
- Gap: No simple system offering instant valuation, doorstep pickup, UPI rewards, and recycling traceability.

KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

Main Difficulties and Bottlenecks

- E-waste disposal is inconvenient, device values are unclear, and consumers have limited access to verified recyclers, easy pickup, and tracking.

Root Causes

- The collection network is fragmented, valuation is mostly manual, informal channels dominate, and there are few simple digital platforms connecting consumers with recyclers.

Consequences

- Old electronics remain unused or may be improperly handled, valuable materials are lost, pollution can increase, and less e-waste reaches formal recycling systems.

EXISTING SOLUTIONS / APPROACHES

What is already being done, and where it falls short

Existing methods, systems or alternatives

- Informal kabadiwalas and scrap dealers: doorstep pickup, pay cash on the spot but unsafe burning and acid leaching, no traceability, no guarantee of safe disposal
- Resale and buyback platforms (Cashify, Amazon/Flipkart exchange, brand trade-ins): instant quote and pickup, but built for working or resellable devices, the focus on resale value, so dead or very old devices and accessories get little or nothing and aren’t tracked to a recycler
- Brand take-back and EPR collection programs :limited collection points, little consumer awareness, few or no rewards
- Authorized recyclers and NGO or municipal drives: safe, certified processing but mostly serve bulk or corporate clients, with few consumer-facing options, and drives are occasional and hard to find

No single platform combines instant valuation, doorstep pickup, UPI rewards, and end-to-end tracking to a verified recycler for all e-waste, including non-working devices

TEAM AETHER

PROBLEM INSIGHTS & OPPORTUNITY AREAS

GENESIS

Resources required to solve the problem

Hardware

- Smartphone or laptop
- internet access
- QR labels for tracking

Software

- Web/mobile-friendly interface
- Database for device and collection status
- Maps, notifications, and reward-status integration
- Recycler verification workflow

People Insights

- Authorized recyclers and pickup partners
- Support for user queries
- E-waste safety and privacy guidance
- software developers

TEAM AETHER

INITIAL DIRECTION & EXPECTED OUTCOME

GENESIS

What Is Your Expected Outcome?

1

USER ENTERS DEVICE

Select device type, condition, age, and location.

2

INSTANT ESTIMATE

Display an indicative value and recycling options.

3

PICKUP / DROP-OFF

Choose a convenient collection method.

4

VERIFIED HANDOVER

Send the device to a verified recycler and update status.

5

REWARD + RECEIPT

Issue reward after confirmation and provide a receipt.